Rishi Pandey

Python Developer (AI & Backend)

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PROFILE

Python Developer (AI & Backend) with 4.5+ years building and scaling production LLM systems. Owns the full delivery cycle — from system design to cloud deployment — across multi-agent RAG pipelines, vector and graph database integration, and FastAPI-based microservices on AWS. Focused on performance, LLM cost-efficiency, and end-to-end observability.

SKILLS

Languages	Python · SQL
Frameworks	FastAPI · Django · Django REST Framework
LLM Systems & RAG Eng.	LangChain · LangGraph (multi-agent orchestration) · RAG Pipelines · Prompt Engineering · Streaming Responses · AWS Bedrock
Observability	Langfuse — distributed tracing · token usage tracking · latency monitoring · LLM cost optimization
Databases	PostgreSQL · pgvector (vector similarity search) · Neo4j (graph DB, Cypher) · MongoDB · Redis (caching & session state)
Infrastructure	Docker · Docker Compose · AWS (EC2, S3, Bedrock) · Kafka · Linux · Git · Shell Scripting
Data & Parsing	Docling (PDF & table extraction) · Selenium · BeautifulSoup · Excel/CSV ingestion pipelines
Engineering Practices	Microservices · RBAC · API Versioning · Pytest · Clean Architecture

EXPERIENCE

Brainerhub Solutions

Python Developer – AI & Backend

Jun 2022 – Present

Architect and own end-to-end delivery of production AI backend systems for enterprise clients — from initial system design through containerized cloud deployment.

- Engineered async FastAPI services integrated with LangChain agents, reducing average API response latency by ~35% via non-blocking I/O, connection pooling, and structured prompt orchestration.
- Designed and shipped multi-agent RAG pipelines in LangGraph with tool-augmented decision trees, dynamic fallback strategies, and retrieval across pgvector (semantic) and Neo4j (relational graph) sources.
- Introduced Langfuse across all LLM workflows — enabling distributed tracing, per-request token accounting, and latency dashboards — which drove a ~20% reduction in inference costs through data-informed prompt optimization.
- Standardized Docker-based deployments across 5+ client projects, cutting environment setup from several hours to under 10 minutes and eliminating environment drift as a source of production incidents.
- Architected Kafka-based event pipelines to decouple high-throughput services, improving processing capacity and fault isolation across data-heavy, multi-service workflows.

Coodeit Solutions

Backend Developer

Dec 2021 – Jun 2022

Owned core API infrastructure and data pipeline design for an AI-integrated product, embedding ML model outputs into production backend services from day one.

- Delivered Django REST Framework APIs serving 1,000+ daily prediction requests, exposing ML inference through clean, versioned contracts with minimal added latency.
- Optimized high-volume PostgreSQL schemas via strategic indexing and query restructuring, improving read performance by ~40% on core production workloads.
- Established shared API contracts with the ML team for model output integration, eliminating interface ambiguity and reducing cross-team debugging cycles.

KEY PROJECTS

AI Risk Matrix Agent

LangChain · LangGraph · FastAPI · AWS Bedrock · Neo4j · pgvector · Redis · Docling · Docker · AWS EC2

- Eliminated a multi-hour manual process by architecting a multi-agent system that generates industrial risk matrices from natural language queries or uploaded images in real time.
- Engineered a dual knowledge base: structured Excel data vectorized into pgvector with rich metadata; unstructured documents parsed via Docling — delivering accurate retrieval across heterogeneous sources.
- Orchestrated agent decision flows in LangGraph with dynamic tool routing and fallback logic; leveraged Neo4j for entity-relationship traversal and Redis for per-session multi-turn memory.
- Neo4j graph retrieval cut query latency by ~50% vs. full-text search; AWS Bedrock handled managed LLM inference at scale, eliminating GPU infrastructure overhead.
- Deployed on AWS EC2 via Docker with per-session state isolation, supporting concurrent users across independent, stateful conversation contexts.

Medical Report RAG Pipeline

FastAPI · LangChain · Docling · pgvector · PostgreSQL · Docker

- Architected a privacy-first RAG system enabling patients to query medical reports in natural language, cutting document review time from 20+ minutes to under 30 seconds.
- Leveraged Docling's table-aware PDF parser to extract structured clinical data with high accuracy on complex layouts where standard extractors failed — removing a critical data quality bottleneck.
- Ran all embedding and LLM inference on client-hosted models with zero third-party data exposure, meeting strict healthcare privacy requirements without compromising retrieval quality.
- Delivered sub-second similarity search across hundreds of reports using pgvector in a Dockerized PostgreSQL instance — no additional search infrastructure required.

Digital Twin Platform

FastAPI · PostgreSQL · SQLAlchemy · Alembic · AWS S3 · Databricks · Pytest · Docker

- Led backend development of an enterprise simulation platform supporting real-time industrial workflow modelling and data-driven operational decisions across multi-tenant environments.
- Engineered CSV/Excel ingestion APIs with dynamic PostgreSQL schema generation, reducing data onboarding effort by ~60% for enterprise clients.
- Designed fine-grained RBAC for multi-tenant access control; achieved 90%+ test coverage via Pytest unit and integration suites across all critical API paths.

Discord Community Bot

Python · PostgreSQL · Docker Compose · Alembic · AWS EC2 · Pillow

- Built a modular, production-deployed Discord bot handling community automation, third-party API integrations, and dynamic image generation via Pillow.
- Maintained 99%+ uptime on AWS EC2 via Docker Compose orchestration; managed schema evolution across all releases using Alembic versioned migrations.

OTHER PROJECTS

Asset Gateway [FastAPI · PostgreSQL · SQLAlchemy 2.0 · Jinja]

- Full-stack inventory management system with dynamic CRUD APIs, SQLAlchemy 2.0 ORM, and server-side Jinja rendering.

Sports News Scraper [Django · Selenium · BeautifulSoup]

- Client-facing Django API for on-demand scraping of JS-rendered sports news; handled dynamic content reliably at scale using Selenium and BeautifulSoup.

EDUCATION

B.Tech – Electrical Engineering

Gyanmanjari Institute of Technology, Bhavnagar · 2018 – 2021

Diploma – Electrical Engineering

Sir Bhavsinhji Polytechnic Institute, Bhavnagar · 2015 – 2018